

Loan Default Prediction — Model Evaluation Report

Comparison of CatBoost, XGBoost, and Random Forest classifiers

1. Executive Summary

Three tree-based classifiers were tuned via GridSearchCV (optimizing F1) to predict loan default on a held-out test set of 7,780 applicants (6,079 non-default / 1,701 default). All three models generalize well, with ROC-AUC between 0.93 and 0.95 and overall accuracy around 90-91%. **XGBoost** achieves the best discrimination (ROC-AUC 0.949, PR-AUC 0.905) and the highest accuracy (91.4%), but carries the highest estimated business cost (279,900). **Random Forest** has slightly weaker ranking metrics but the **lowest business cost (235,410)**, making it the most attractive option once the asymmetric cost of false positives vs. false negatives is taken into account. **CatBoost** sits in between on every metric and shows a moderately good calibration.

2. Model Comparison

Metric	CatBoost	XGBoost	Random Forest
Accuracy	90.5%	91.4%	90.3%
ROC-AUC	0.940	0.949	0.934
PR-AUC	0.893	0.905	0.886
Brier score	0.0598	0.0561	0.0601
Business cost (€)	264,226	279,900	235,410
Decision threshold	0.31	0.31	0.28
Precision / Recall (Default)	0.77 / 0.80	0.80 / 0.80	0.77 / 0.79
False Negatives / False Positive	332 / 410	333 / 339	352 / 400
Key hyperparameters	depth=8, iterations=150, lr=0.1, l2_leaf_reg=1, class_weight={0:1,1:2}	max_depth=6, n_estimators=150, lr=0.1, reg_lambda=1, scale_pos_weight=2	max_depth=20, n_estimators=300, min_samples_leaf=2, class_weight={0:1,1:2}

3. Recommendation

If the objective is **pure predictive discrimination**, **XGBoost** is the strongest model (highest ROC-AUC, PR-AUC and accuracy). If the objective is **minimizing real-world financial loss** — the more business-relevant criterion — **Random Forest** is preferable, since its lower operating threshold (0.28) and error mix translate into the lowest total cost despite marginally lower ranking metrics. CatBoost is a reasonable, well-calibrated middle ground and its calibration curve (Section 5) tracks

the ideal line more closely than the other two models in the mid-probability range.

4. Feature Importance & Explainability

Across the Random Forest model, **Loan_percent_income**, **internal_credit_rating**, **Income** and **Interest_rate** are by far the strongest predictors of default, together accounting for the majority of total importance. Engineered ratio features and home-ownership status add secondary signal, while loan-intent and age-group dummies contribute very little. A SHAP explanation for one borderline example confirms the same drivers at the individual level: loan-to-income burden and renting push risk up, offset by a favorable credit rating.

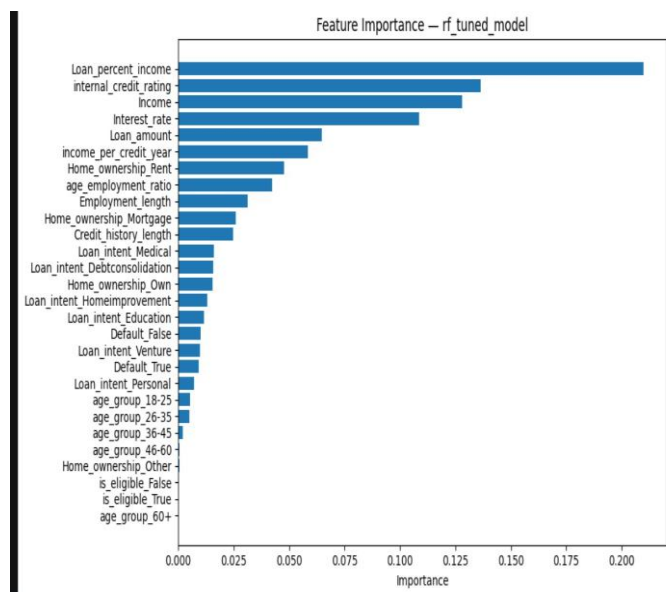


Figure 1 — Global feature importance (Random Forest).

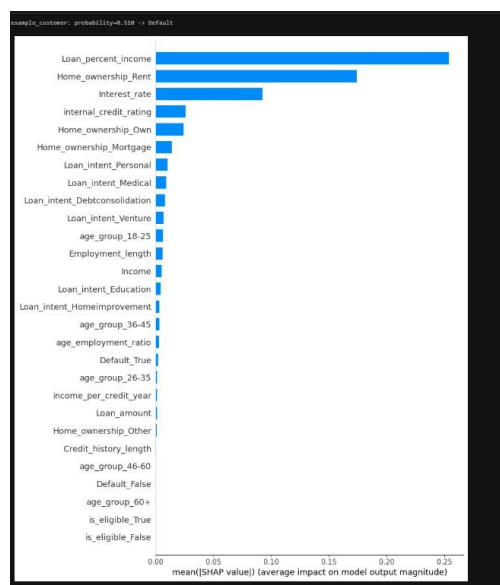
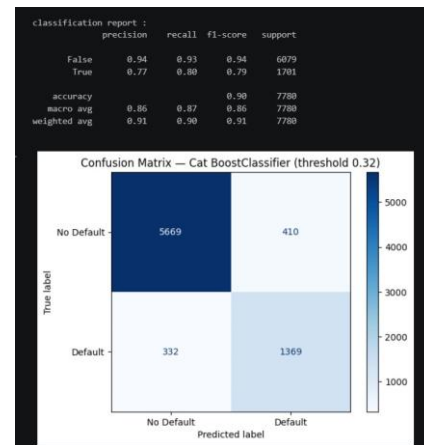
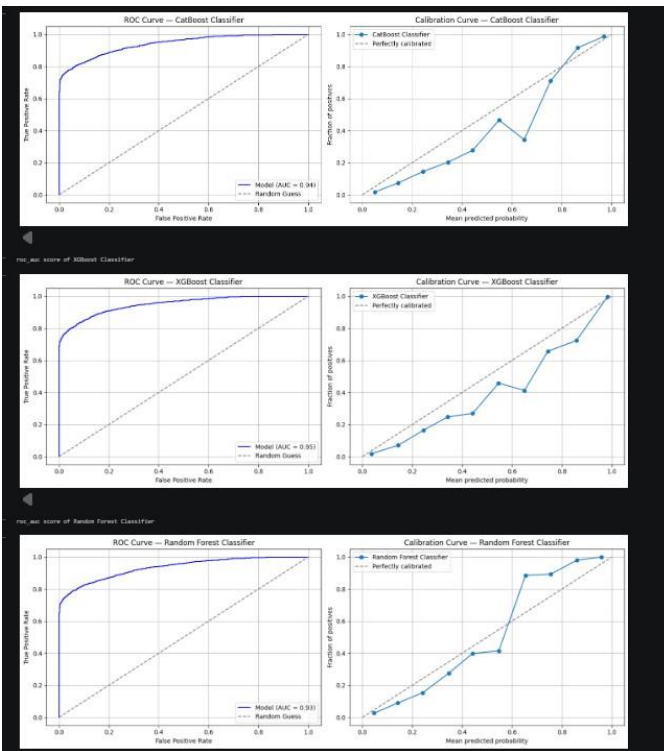


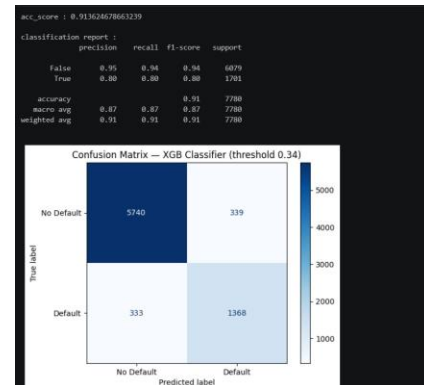
Figure 2 — SHAP for one example customer.

5. Discrimination, Calibration & Confusion Matrices

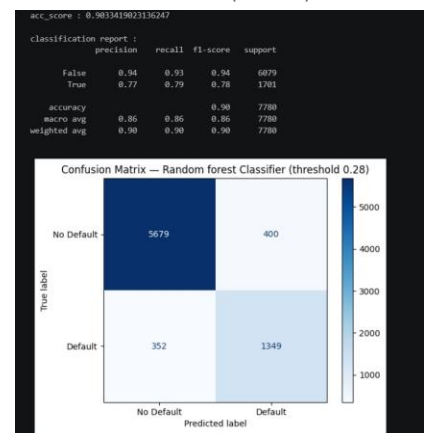
All three models show strong separation between defaulters and non-defaulters (ROC-AUC 0.93-0.95). Calibration is imperfect for all in the 0.4-0.7 predicted-probability band — each curve dips below the diagonal there, meaning predicted probabilities overstate true risk in that range — but XGBoost tracks the ideal line most closely overall.



CatBoost (thr 0.32)



XGBoost (thr 0.34)



Random Forest (thr 0.28)

Figure 3 (left) — ROC & calibration curves, top to bottom: CatBoost, XGBoost, Random Forest. Figure 4 (right) — confusion matrices at each model's tuned threshold.

6. Conclusion

All three tuned models are viable for production, comfortably clearing 90% accuracy and ~0.93+ ROC-AUC on unseen data. Model selection should be driven by business priorities: I chose Random forest over Xgboost to minimize expected losses from misclassified applicants. CatBoost offers a balanced, well-calibrated alternative if governance or interpretability tooling favors it.